Numerical Analysis Assignments

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Contents

1. Chapter 4 assignment
2. Chapter 5 assignment

Chapter 4 assignment

1. Find the Lagrange-form interpolating polynomial of this table:

x	0	1	4
y	1	2	2

2. Find the Newton-form interpolating polynomial of the above table.

3. Construct the natural cubic spline that interpolates

x	0	1	2	3	4
y	1	2	0	3	1

Give detailed steps.

Chapter 5 assignment

1. Apply the composite trapezoid rule, with $h = \frac{1}{4}$, to approximate the integral $\int_0^1 \frac{1}{1+x^3} dx$. Its exact value is $\frac{1}{3}\ln 2 + \frac{1}{9}\sqrt{3}\pi$.
2. Apply composite Simpson's rule, with $h = \frac{1}{4}$, to approximate the integral $\int_0^1 \frac{1}{1+x^3} dx$.
3. Read the lecture notes on composite quadrature, and derive the composite Boole's rule.